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THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Mirage Kitten Aviation and FinTech Espionage Malware Campaign Daily Priority Update

Threat Report

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TLP:CLEAR | Lost Boys Cyber | Mirage Kitten Aviation and FinTech Espionage Malware Campaign Daily Priority Update - Threat Report

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1. Executive Summary

Kaspersky/Securelist published reporting on 2026-09-01 describing Mirage Kitten activity targeting aviation and FinTech sectors across the Middle East and Africa with previously undocumented Node.js and JavaScript malware families identified as NodeRabbit and PollCat. This follows earlier Securelist reporting in late July 2026 on Mirage Kitten tooling including NightLedger, BridgeHead, and ArcBridge.

This report is article-backed threat intelligence, not a sample-backed malware-analysis report. No TRASHcan detonation, Ghidra reverse engineering, or local sample execution was performed in this cron run. The defender value is therefore campaign triage: identify potentially exposed aviation/aerospace and FinTech environments, monitor for script-runtime abuse, review long-term access paths, and prioritize telemetry that can surface Node.js/JavaScript malware execution.

Aviation/aerospace operators should treat this as a priority because the reported victimology directly includes aviation, and because airline, airport, MRO, aerospace, and defense-adjacent organizations frequently expose supplier portals, remote-access paths, and high-value document repositories that align with espionage objectives.

2. Intelligence Requirements

Question	Current Answer	Confidence
Does the reporting explicitly mention aviation?	Yes. Securelist headline and metadata identify aviation and FinTech targeting across the Middle East and Africa.	High
Are malware families named?	Yes. NodeRabbit in Node.js and PollCat in JavaScript are described in public metadata; earlier related reporting named NightLedger, BridgeHead, and ArcBridge.	High
Is local malware evidence available?	No uploaded sample or sandbox telemetry was available to this scheduled run.	High

3. Source Review & Confidence

Source	Contribution	Confidence
Securelist / Kaspersky 2026-09-01	Primary current reporting naming Mirage Kitten, aviation/FinTech targeting, NodeRabbit, and PollCat	High
Securelist / Kaspersky 2026-07-29	Related prior reporting on Mirage Kitten new tools and espionage campaign context	High
Regional/security press coverage	Corroborates sector and regional targeting claims from Kaspersky disclosures	Medium
Public CTI naming context	Supports actor-alias awareness but not used as proof of new activity	Low-Medium

4. Threat Activity Overview

Mirage Kitten is described in public reporting as an espionage-focused actor set with activity across the Middle East and Africa. The current Securelist article states that Kaspersky researchers discovered attacks using previously undocumented malware families: NodeRabbit, implemented with Node.js, and PollCat, implemented in JavaScript.

Because the available evidence is vendor reporting rather than local sample telemetry, this report focuses on observable tradecraft: script-runtime execution, long-term access, tunneling or backdoor behavior described in related reporting, and targeting of aviation/FinTech organizations.

5. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Likely Exposure Paths	Defender Action
Airlines	Reservation, crew, airport operations, and supplier portals hold high-value identity/document data.	Phishing, compromised web portals, vendor remote access, endpoint compromise.	Monitor script-runtime execution and third-party access paths.
Airports	Operational networks and tenant ecosystems create broad trust boundaries.	Contractor endpoints, shared support systems, unmanaged browsing from operations teams.	Segment operational systems and monitor abnormal egress.
MRO / aerospace suppliers	Engineering documents and maintenance records are espionage-relevant.	Supplier portals, VPN, file-transfer systems, developer/engineering workstations.	Harden identity, inspect Node.js/JavaScript execution on endpoints.
Defense-adjacent aerospace	Overlaps with national-security collection priorities.	Spearphishing, credential theft, supplier compromise.	Prioritize executive/engineering identity protection and DLP telemetry.

6. Affected Sectors and Exposure

The primary current reporting names aviation and FinTech across the Middle East and Africa. Related public reporting mentions aerospace, aviation, defense, and telecommunications as historically relevant sectors for this actor/toolset. Organizations outside those regions should still review exposure if they have regional subsidiaries, suppliers, codeshare partners, MRO relationships, or shared identity infrastructure.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	T1566 Phishing	Espionage campaigns commonly rely on targeted delivery; validate against local email/web telemetry.
Execution	T1059.007 JavaScript / T1059 command scripting	Current reporting names JavaScript malware PollCat and Node.js malware NodeRabbit.
Persistence	T1105 Ingress Tool Transfer / backdoor staging	Earlier reporting names backdoors/tunnelers; current report names malware families used for access.
Command and Control	T1090 Proxy / tunneling	Prior Mirage Kitten reporting named

		BridgeHead/ArcBridge tunneling tools.
Collection	T1005 Data from Local System	Espionage objective implies collection from compromised endpoints; confirm locally before incident attribution.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
NodeRabbit	Malware family / Node.js payload	Hunt for Node runtime execution outside developer baselines.
PollCat	Malware family / JavaScript payload	Hunt for suspicious wscript/cscript/node execution chains.
NightLedger	Backdoor named in related reporting	Context for Mirage Kitten tooling; not asserted as present in every intrusion.
BridgeHead / ArcBridge	Tunneling tools named in related reporting	Hunt for unauthorized tunnel behavior and unusual egress.
Mirage Kitten / UNC1549 / Smoke Sandstorm / Nimbus Manticore naming	Actor/cluster aliases	Use for intelligence correlation, not as automatic attribution proof.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Node.js execution from non-developer paths	EDR process telemetry	Surface NodeRabbit-like runtime abuse
Windows Script Host running recently written JavaScript	EDR/Sysmon	Surface PollCat-like execution
Script runtime initiating outbound connections	EDR/network telemetry	Detect backdoor or staging behavior
Tunnel-like long-lived outbound sessions from aviation/engineering endpoints	Proxy/firewall/EDR	Find unauthorized remote access channels

9.2. Sigma / Detection Content

```

title: Suspicious Node.js or JavaScript Runtime Execution in Aviation Environments
status: experimental
description: Hunts for script-runtime behavior relevant to Mirage Kitten reporting on NodeRabbit and PollCat. Tune developer and admin baselines before alerting broadly.
logsource:
  product: windows
  category: process_creation
detection:
  script_runtime:
    Image|endswith:
      - '\node.exe'
      - '\wscript.exe'
      - '\cscript.exe'
  suspicious_path:

```

```

CommandLine|contains:
  - '\AppData\'
  - '\ProgramData\'
  - '\Temp\'
  - '.js'
network_tooling_hint:
  CommandLine|contains:
    - 'http'
    - 'ws://'
    - 'wss://'
    - 'socket'
condition: script_runtime and suspicious_path
fields:
  - UtcTime
  - User
  - Image
  - CommandLine
  - ParentImage
falsepositives:
  - Developer tooling, legitimate Node.js applications, software installers
level: medium

```

9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(30d)
| where FileName in~ ('node.exe', 'wscript.exe', 'cscript.exe', 'mshta.exe')
| where ProcessCommandLine has_any ('AppData', 'ProgramData', 'Temp', '.js', 'http', 'wss://', 'socket')
| summarize ExecCount=count(), FirstSeen=min(Timestamp), LastSeen=max(Timestamp),
Users=dcount(AccountName) by DeviceName, FileName, InitiatingProcessFileName, ProcessCommandLine
| order by ExecCount desc

```

10. Threat Hunting

10.1. Hunt Hypothesis

Aviation or aerospace endpoints impacted by Mirage Kitten-style activity will show script-runtime execution from user-writable locations followed by network connections to unfamiliar external infrastructure, with a bias toward engineering, finance, executive, travel, or supplier-access users.

10.2. Hunt Query

```

let ScriptRuns = DeviceProcessEvents
| where Timestamp > ago(30d)
| where FileName in~ ('node.exe', 'wscript.exe', 'cscript.exe', 'mshta.exe')
| where FolderPath has_any ('AppData', 'ProgramData', 'Temp') or ProcessCommandLine has_any ('.js', 'http', 'wss://');
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where InitiatingProcessFileName in~ ('node.exe', 'wscript.exe', 'cscript.exe', 'mshta.exe')
| join kind=innerunique ScriptRuns on DeviceId
| where Timestamp between (Timestamp1 .. datetime_add('hour', 4, Timestamp1))
| summarize Connections=count(), RemoteIPs=make_set(RemoteIP, 20), RemoteUrls=make_set(RemoteUrl, 20),
FirstSeen=min(Timestamp), LastSeen=max(Timestamp) by DeviceName, AccountName,
InitiatingProcessFileName, InitiatingProcessCommandLine
| order by Connections desc

```

11. Risk Assessment

Dimension	Assessment
Likelihood	Medium-high for organizations in named sectors/regions; medium elsewhere with regional or supplier links.
Impact	High for aviation/aerospace due to espionage, operational

	planning, engineering data, and third-party trust relationships.
Confidence	High that Kaspersky reported the activity; medium for environment-specific exposure without local telemetry.

12. Recommended Actions

Priority	Owner	Action
Critical	Aviation/aerospace SOC	Deploy script-runtime hunts against aviation, engineering, finance, and supplier-access endpoints.
High	Identity team	Review MFA and conditional access for regional subsidiaries, third-party VPN, and supplier portals.
High	Endpoint engineering	Restrict Windows Script Host and unmanaged Node.js execution where business-feasible.
Medium	Threat intel	Track Mirage Kitten aliases and update watchlists as Kaspersky publishes additional IOCs.
Medium	Network security	Review egress controls for tunnel-like traffic from non-developer endpoints.

13. References

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